

PHYS 253: Thermal Physics

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Abstract

NOT FINISHED

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1 Introduction

Textbook used is An Introduction to Thermal Physics by Daniel V. Schroeder. Taught by Bradley Siwick. Mondays, Wednesdays and Fridays from 8:35 to 9:25 in rutherford 118. Course grading is divided as follows:

- 10% problem sets (~ 6 , must be handwritten pdf scans) ¹.
- 50% 2 midterms.
- 40% final exam.

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Definition 1 (Thermodynamics). *Thermodynamics, derived from the ancient Greek words *therme* (heat) and *dynamis* (power), is the branch of physics that deals with the relationships between heat, work, temperature, and energy. It is the study of equilibrium properties of large-scale systems in which temperature is an important variable.*

Remark 1. *Thermodynamics does not give microscopic details of the system², but rather describes the macroscopic properties and behaviors of systems in terms of a few state variables.*

2 Thermal Equilibrium

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2.1 Equilibrium States and State Variables

Definition 2 (Equilibrium state). *A system is in ES if all the measurable bulk physical properties³ of the system are uniform throughout the system and in time.*

Remark 2. *If the system is a fixed volume of gas, the pressure is uniform and volume is constant in time in an equilibrium state.*

Remark 3. *The pressure, volume and mass of a gas fully describe the equilibrium state of the gas, and fixes all the other measurable bulk properties.*

2.2 Equations of State

Definition 3 (Equation of state). *An equation of state is a functional relationship between the easily measured state variables⁴ of a system in an equilibrium state.*

¹Only one or two of the problems on each assignment will be indicated for submission and grading.

²Classical thermodynamics cannot describe microscopic systems directly, but modern extensions like stochastic thermodynamics and statistical mechanics can.

³Could be any mechanical and structural properties, thermal, electrical, optical...

⁴Such as pressure, volume, and temperature

Example. *The ideal gas law is an example of an equation of state that relates (P, V, T) and applies to equilibrium states of dilute gases at high temperature: $PV=nRT$, where R is the ideal gas constant.*

Definition 4 (State function). *A state function is a property of a system that depends only on the current equilibrium state of the system, and not on the path taken to reach that state. Examples of state functions include internal energy, enthalpy, entropy, and Gibbs free energy.*

Remark 4. *State functions are important in thermodynamics since they do not depend on how the equilibrium was reached.*

2.3 Temperature and the Zeroth Law

Temperature is a state function, and is defined as the property that determines whether two systems are in thermal equilibrium⁵ with each other.

Law 1 (Zeroth Law of Thermodynamics). *If two systems (A & B) are separately in thermal equilibrium with a third system (C), meaning no heat flows between A and C , and no heat flows between B and C , then they are in thermal equilibrium with each other.*

3 Misc

A Appendix

B Useful Links

⁵Not in thermodynamic equilibrium!